

Pushkar Jajoria

Curriculum Vitae

Spoken Language Systems (LSV)

Saarland University

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🌐 My Webpage

🐙 Github in LinkedIn

Education

- 2019–2021 : **Master of Informatics, Artificial Intelligence**, *Università della Svizzera Italiana*, Lugano, Switzerland.
- 2010–2014 : **Bachelor of Computer Science & Engineering**, *Indian Institute of Technology*, Hyderabad.

Publications

In Conference Proceedings

- 2026 **Jajoria, Pushkar**, Arianna Graciotti, Giovanna Casali, Jesujoba O. Alabi, Rodolfo Delmonte, Angelo Pompilio, Rocco Tripodi, James McDermott, and Dietrich Klakow. Audio-lyrics alignment dataset for italian arias. In *Proceedings of the 15th Language Resources and Evaluation Conference (LREC 2026)*, Palma de Mallorca, May 2026. ELRA Language Resources Association, May 2026.
- 2025 **Jajoria, Pushkar**, Dietrich Klakow, and James McDermott. Text conditioned symbolic drumbeat generation using latent diffusion models. In *Proceedings of the 22nd Sound and Music Computing Conference (SMC2025)*, Graz, July 2025, pages 311–318. Institute of Electronic Music and Acoustics, University of Music and Performing Arts Graz, July 2025.
- 2024 **Jajoria, Pushkar**, Aurel Mäder, Florian Krenn, and James McDermott. Prevalence of tresillo rhythm in contemporary popular music. In *Proceedings of the 21st Sound and Music Computing Conference (SMC2024)*, Porto, July 2024. Sound and Music Computing Conference, July 2024.
- 2024 Rory Sweeney, **Jajoria, Pushkar**, Danny Diamond, Mathieu D'Aquin, and James McDermott. Patterns ui: An interactive tool for music exploration. *Perspectives on Immersion through Laser Doppler Vibrometry*, page 65, 2024.
- 2024 Abdul Shahid, Rory Sweeney, **Jajoria, Pushkar**, Danny Diamond, Mathieu D'Aquin, and James McDermott. Ontologies, knowledge graphs, and user interfaces for exploration of irish traditional music. 2024.

Research Experience

Saarland University

- 2026 – present **Molecular Property Prediction via Chemical Language Models.**
Investigating whether learned structural tokens from frozen CLMs (MoLFormer, ChemBERTa) can improve LLM performance on molecular tasks through a trainable Q-Former bridge module.
- 2025 - 2025 **Anti-Inflammatory Drug Discovery.**
Using domain adaption on foundational chemical language models to predict anti-inflammatory properties of new chemicals.
- 2024 - 2025 **Text-conditioned Symbolic Drumbeat Generation.**
Generating symbolic drumbeat conditioned on text prompts using latent diffusion models.
- Advisor : **Prof. Dietrich Klakow**, LSV, Saarland University

University of Galway, Ireland

2022 – 2023 **Visualizing knowledge graphs for musical patterns in Irish traditional music..**
Developing tools to visualize musical patterns in Irish folk music and linking them with other songs in the corpora.

2022 – 2023 **Prevalence of the Tresillo rhythm in popular music.**
Measuring the use of Tresillo rhythm in US Billboard Top-20 Charts across 20 years.

Advisor : **Prof. James McDermott**, University of Galway

Università della Svizzera italiana & École polytechnique fédérale de Lausanne

2020 – 2021 **Sequential prediction of polyphonic music using human interpretable features.**
Developed an interpretable model for polyphonic music prediction using feature-based policy learning and nested Markov processes, achieving over 90% accuracy on Bach chorales and demonstrating learned principles of voice leading and harmony.

Advisor : **Prof. Martin Rohrmeier**, DCML, EPFL and **Prof. Cesare Alippi**, Università della Svizzera italiana

Professional Experience

2024 – present **Saarland University**, PhD Researcher, Spoken Language Systems (LSV).

2022 – 2024 **University of Galway**, Research Assistant.

2018 – 2019 **Delhivery**, Senior Developer. Invited to give an internal talk on *AI for warehouse management*.

2016 – 2018 **Joveo**, Software Developer.

2014 – 2015 **Salesforce**, Associate Member of Technical Staff.

Achievements & Recognitions

2015 **Award** for outstanding project in Salesforce Hackathon.

2019 Invited to give a **talk** on *AI for warehouse management* at Delhivery.

2020 **Outstanding project** of Data Science at EPFL for delay aware journey planner for public transport in Zurich.

Computer skills

Programming Languages Python, PyTorch, Tensorflow

Teaching & Service

2025 **Seminar on Graph Neural Networks for Molecular Property Prediction**, Saarland University.